Hochschule Fulda
University of Applied Sciences

Global Distributed Software Development/
Teamprojekt (AI5088)

“Master Team Project Fall 2024”

Way2Home

Milestone 1
(Dated: 11/12/2024)

Supervisor: Prof. Dr. Todtenhöfer, Rainer

Team Members and roles

Member Name	Email	Roles
Romina Nikasa	romina.nikasa@informatik.hs-fulda.de	Team Lead
Rashid Hussain Mohammed	rashid-hussain.mohammed@informatik.hs-fulda.de	GitHub Master
Tim Schröder	tim.schroeder@informatik.hs-fulda.de	Backend Lead
Anmoldeep Angadi	anmoldeep.angadi-manjunatha@informatik.hs-fulda.de	Frontend Lead
Samarth Mahajan	samarth-omkar.mahajan@informatik.hs-fulda.de	Frontend Developer
Suhaila Kondappilly Aliyar	suhaila.kondappilly-aliyar@informatik.hs-fulda.de	Backend Developer
Maira Shafi	maira.shafi@informatik.hs-fulda.de	Documentation Master

Revision History

Date Submitted	Date Revised	Revision Summary	Version
13/11/2024			

Table of Contents

1. Executive Summary **3**
2. Personas and Main Use Cases **4**
Personas 4
Use Cases 6
UC-1: Guest Views Basic Listings 6
UC-2: Student Filters Listings 6
UC-3: Student Adds Listing to Favorites 6
UC-4: Student Views Nearby Amenities 7
UC-5: Landlord Creates a New Listing 7
UC-6: Moderator Reviews and Approves Listing 7
UC-7: Student Contacts Landlord via In-App Messaging 7
UC-8: Landlord Manages Listings and Responds to Inquiries 7
UC-9: Student Sets Housing Alert Notifications 7
UC-10: Moderator Addresses a User Complaint 7
UC-11: Admin Monitors Platform Activity and Security 8
3. List of Main Data Items and Entities **8**
4. Initial List of Functional Requirements **10**
5. List of Non-Functional Requirements **13**
6. Competitive Analysis **14**
7. High-Level System Architecture and Technologies Used **15**
Technologies 15
Deployment info 16
8. Team and Roles **16**
9. Checklist **17**

1. **Executive Summary**

Our team is developing a secure, user-centered web platform specifically designed to assist Fulda University students in finding reliable apartment and housing rentals. Recognizing the challenges students—especially international ones—face in the current rental market, our platform offers a safe, moderated environment with features tailored to their needs. Importantly, our team includes Fulda University students, including international members with firsthand experience with existing systems' frustrations. This personal insight enables us to design a platform that authentically addresses student expectations.

Key Features

- **Advanced Search & Filtering:** Refine searches by price, location, pet-friendliness, accessibility, and neighborhood traits.
- **Easy Property Posting:** Quick listing process for both student and non-student landlords.
- **Secure, Verified Listings:** Only admin-approved listings are visible, ensuring quality and compliance.
- **In-Site Messaging:** Enables private, secure communication between students and landlords.
- **Housing Alerts:** Personalized alerts notify users of new listings matching their criteria.
- **Responsive UI:** Optimized for both web and mobile, ensuring a seamless experience on any device.
- **Media-Rich Listings:** High-quality images and videos make listings informative and visually engaging.
- **Admin Moderation:** Backend moderation ensures adherence to platform guidelines and legal standards.
- **Contact Support:** The “Contact Us” form provides easy access to support for user inquiries.
- **Favorites List:** Allows students to save listings to a "Favorites" list, enabling easy comparison and later access without reapplying search filters.
- **Nearby Amenities View:** Students can assess neighborhood suitability by viewing nearby amenities like cafes, transport hubs, and other points of interest.
- **Tenant Score Based on Landlord Questionnaire:** Each student receives a "Tenant Score" based on responses to a landlord-specific questionnaire. The score is visible only to the landlord who created the questionnaire, aiding in sorting and filtering potential tenants who align with their preferences.
- **Virtual Property Tours:** 360-degree virtual tours offer remote viewing, ideal for international students.
- **Video Call Request:** Students and landlords can request and schedule video calls, facilitating direct communication on rental details.

Key Advantages, Novelty, and Value

Way2Home stands out for its focus on verified listings and user safety, setting it apart from other rental platforms. Unique features like the “Tenant Score” and “Tailored Housing Alerts” offer tailored support to students, especially those unable to visit properties in person. Our team’s deep understanding of student needs, combined with technical expertise, enables us to deliver a high-quality, secure rental experience.

Why Fund Way2Home

Supporting Way2Home means investing in a vital solution to Fulda University’s housing challenges. With funding, we can enhance security, expand features, and scale our reach, helping more students navigate their housing search safely and effectively. This investment not only simplifies housing for students but fosters a secure and trusted rental environment at a time of critical need.

2. Personas and Main Use Cases

Personas

Persona 1: Emily Li – The Student User (Primary Persona)

- **Characteristics**: Emily is a 20-year-old undergraduate international student, new to the area, looking for affordable, conveniently located housing close to campus. Balancing full-time studies, she values a streamlined experience that allows her to quickly find safe, reputable housing options.
- **Goals**: Quickly locate budget-friendly housing close to campus or transit; easily shortlist and contact landlords; save potential rentals to review later.
- **Skills**: Proficient with mobile and web platforms, accustomed to using filters and search functions, and comfortable with direct communication through in-app messaging.
- **Pain Points**: Time constraints due to academic demands; budget limitations; unreliable or misleading listings; occasional communication delays with landlords.

Persona 2: Michael Reyes – The Guest User

- **Characteristics**: Michael is a parent visiting from out of town, helping his daughter explore potential housing options. He’s unfamiliar with the local rental market but wants an overview of availability and pricing.
- **Goals**: Browse current listings, gain an understanding of pricing, property types, and amenities; assist his daughter in finding suitable options.

- **Skills**: Basic web navigation, interested primarily in browsing without the need for interactive features.
- **Pain Points**: Limited access to property details and contact information as a guest; may need more functionality to explore listings in-depth, which could reduce initial engagement.

Persona 3: Linda Weber – The Landlord

- **Characteristics**: Linda owns a small property management business, renting out several apartments near the university. She values efficiency and relies on digital platforms to reach student tenants.
- **Goals**: Quickly create and manage listings, attract responsible tenants, and facilitate prompt, secure communication with interested students.
- **Skills**: Familiar with online platforms, comfortable managing listings, uploading images, and responding to inquiries, though not deeply technical.
- **Pain Points**: Time-consuming listing creation; challenges in ensuring listings are appealing to students; managing multiple listings and tenant communication effectively.

Persona 4: Alex Jansen – The Admin

- **Characteristics**: Alex is a Fulda University administrator with oversight of the platform, responsible for ensuring quality listings and compliance with university standards.
- **Goals**: Manage website functionality, oversee listing quality, ensure smooth operation and security.
- **Skills**: Skilled in platform administration, data handling, user management, and troubleshooting.
- **Pain Points**: Balancing content moderation with user satisfaction, overseeing a large volume of listings, maintaining security and platform integrity.

Persona 5: Julia Braun – The Moderator

- **Characteristics**: Julia works part-time as a moderator, tasked with reviewing content and ensuring listing quality. With experience in the local rental market, she understands students' needs well.
- **Goals**: Approve or reject listings, maintain listing quality, address user-reported issues or complaints.
- **Skills**: Familiar with web content management, detail-oriented, experienced in review and approval processes.
- **Pain Points**: Ensuring high standards for listings, handling flagged or inappropriate content, maintaining consistency in approvals.

Use Cases

UC-1: Guest Views Basic Listings

- **Actor**: Michael Reyes (Guest)
- **Goal**: Explore basic housing information without an account.
- **Description**: Michael, as a guest, accesses the listings page and views basic property details like rent, location, and apartment photos. Since he doesn't have an account, he cannot contact landlords or save listings.

UC-2: Student Filters Listings

- **Actor**: Emily Li (Student)
- **Goal**: Narrow down property search based on specific criteria.
- **Description**: Emily logs in, applies filters for location, rent, and furnished options, and reviews the listings that match her criteria.

UC-3: Student Adds Listing to Favorites

- **Actor**: Emily Li (Student)
- **Goal**: Save listings for easy comparison and later access.
- **Description**: Emily saves specific listings to her "Favorites" list, allowing her to review and compare them later without reapplying her search filters.

UC-4: Student Views Nearby Amenities

- **Actor**: Emily Li (Student)
- **Goal**: Assess neighborhood suitability based on nearby amenities.
- **Description**: Emily opens the "Nearby Amenities" feature for a listing to check the proximity to cafes, transport hubs, and other points of interest.

UC-5: Landlord Creates a New Listing

- **Actor**: Linda Weber (Landlord)
- **Goal**: Post a new property for rent.
- **Description**: Linda logs in, fills out listing details, uploads photos, and submits the listing for moderator review.

UC-6: Moderator Reviews and Approves Listing

- **Actor**: Julia Braun (Moderator)
- **Goal**: Ensure listing quality and compliance.

- **Description**: Julia reviews Linda's submitted listing, checks for clarity and quality, and provides feedback if needed. Once updated, Julia approves the listing, making it visible on the platform.

UC-7: Student Contacts Landlord via In-App Messaging

- **Actor**: Emily Li (Student)
- **Goal**: Inquire about a specific property.
- **Description**: Emily uses the secure in-app messaging system to ask Linda questions about the property, such as lease length and pet policy.

UC-8: Landlord Manages Listings and Responds to Inquiries

- **Actor**: Linda Weber (Landlord)
- **Goal**: Respond to potential tenant inquiries.
- **Description**: Linda checks her dashboard for messages on her active listings and responds to student inquiries, keeping all communications in one place.

UC-9: Student Sets Housing Alert Notifications

- **Actor**: Emily Li (Student)
- **Goal**: Receive alerts for new listings matching criteria.
- **Description**: Emily enables “Housing Alerts” to notify her when new listings within her budget and preferred location are posted.

UC-10: Moderator Addresses a User Complaint

- **Actor**: Julia Braun (Moderator)
- **Goal**: Investigate and resolve a user-reported issue.
- **Description**: Julia reviews a complaint from a student about a misleading listing and takes appropriate action after verifying details with the landlord.

UC-11: Admin Monitors Platform Activity and Security

- **Actor**: Alex Jansen (Admin)
- **Goal**: Ensure platform integrity and security.
- **Description**: Alex checks site analytics, reviews flagged content, and addresses security concerns to maintain compliance and trustworthiness.

3. List of Main Data Items and Entities

Index	Entity	Description
1	Admin	A User role with permissions for platform management, overseeing flagged content, and handling escalated issues.
2	Moderator	A User role with permissions for listing moderation, reviewing flagged content, and resolving user complaints.
3	Student	A User role with capabilities like saving listings, setting search alerts, messaging, and accessing reviews.
4	Landlord	A User role for creating, managing, and editing property listings, responding to messages, and uploading rental agreements.
5	Guest	Users without an account who can browse property listings but cannot access messaging or favoriting.
6	User Authentication	Manages secure user login and access permissions for Students, Landlords, Guests, Admins, and Moderators. Supports university email verification for Students.
7	Property	Represents a rental property, including essential details such as rent, location, category, amenities, and media assets. Tracks the status of the listing (e.g., draft, published) and links to the landlord's information for management and contact.
8	Listing Categories	Organizes properties by type (e.g., Apartment, Shared Housing) to support advanced search filters for students with specific housing needs.
9	Approval Status	Tracks the current moderation status (pending, approved, rejected) for each property. Managed by moderators.
10	Media Assets	Stores images, videos, and virtual tours related to each Property.
11	Search and Filter Preferences	Stores search filters and alert criteria for students. Generates notifications when new listings match these preferences.
12	Saved Listings	Stores a student's favorite properties for easy access and later review.

13	Message	Stores in-app messages between students and landlords. Includes sender, recipient, message content, and timestamp.
14	Reviews and Ratings	Enables students to leave ratings and reviews for landlords, promoting transparency and helping future renters.
15	Document Exchange	Facilitates the secure upload and exchange of rental agreements and other documents between students and landlords.
16	Tenant Score	Calculates a student's reliability score based on rental history, feedback, and punctuality, visible to landlords.
17	Notification	Manages alerts for messages, listing updates, saved listing activity, and new listings matching student alerts.
18	Flagged Content	Tracks flagged listings or messages for review by moderators. Ensures platform safety and compliance.
19	Virtual Property Tour	Stores 360-degree property tours, allowing students to remotely view property interiors.
20	Video Call Request	Facilitates requests for video calls between students and landlords, including scheduling details and links to the call interface.
21	Rental Application	Allows students to submit rental applications with required documents (e.g., proof of income). Landlords can review and approve or reject applications.
22	Share Property	Enables sharing listings externally (social media, email, etc.).
23	Downloadable Lease Terms	Allows downloadable rental agreement documents post-transaction for students.
24	User Dashboard	A personalized page for users to view saved listings, notifications, interactions, and drafts.
25	Feedback & Support	Manages user feedback, bug reports, and support requests to improve platform responsiveness.
26	Privacy Settings	Enables users to control profile visibility, including who can see contact information.

4. Initial List of Functional Requirements

High-Level Functions: List about 20 functional requirements that describe what the application will do, not how it will be done. Format: Number each requirement uniquely and provide a brief 1-3 line description.

1. **User Profile Management**

Users can complete or update their profiles with contact information, profile photos, and short bios. This helps personalize the user experience and builds credibility, especially for landlords interacting with potential tenants.

2. **Creation of Property Listing**

Landlords can create detailed property listings by adding information such as rent, location, photos, property type, and amenities. This ensures that students have the relevant details needed to evaluate a property without additional inquiries.

3. **Editing of Property Listing**

Landlords can update existing listings to correct information or reflect changes, such as rent adjustments or new photos. This helps keep listings accurate and current, reducing misunderstandings with prospective renters.

4. **Basic Search Functionality**

The system shall allow students to perform a basic search by entering keywords (e.g., property name, location) in a search bar, retrieving relevant property listings based on keyword matches.

5. **Advanced Search and Filtering**

Students can filter search results based on various criteria, like price, location, pet-friendliness, and accessibility. This feature makes it easy for users to find properties that match their specific needs, saving time and improving their experience.

6. **View Detailed Property Information**

Students & guests can view a property's full details, including high-quality images, rent, neighborhood characteristics, and landlord information. This reduces the need for initial inquiries and provides students with a complete picture of what each property offers.

7. **In-Site Messaging System**

A secure, internal messaging system allows students and landlords to communicate directly on the platform. They can discuss property details, rental terms, and arrange viewings without relying on external communication, enhancing privacy and convenience.

8. **Admin Oversight of Platform Analytics**

The system shall provide admins with access to site analytics, including user activity, login attempts, and page views, to monitor usage trends and detect unusual patterns. Additionally, it shall allow admins to review and assess flagged content from moderators.

9. **Moderator Reviewing and Approving Listing**

The system shall enable moderators to review and approve property listings and user-generated content, ensuring compliance with platform guidelines before publication.

10. **Encountering Inappropriate Listings**

If students encounter inappropriate or misleading listings, they can report them to the moderator. This feedback is essential for maintaining a safe and accurate platform, as moderators can investigate reports and take corrective actions when needed.

11. **Favouriting Listings**

Students can save listings of interest to revisit later, making it easy to keep track of potential rentals without performing multiple searches. This feature enhances convenience and helps students organize their property options.

12. **Creating a Draft Listing**

Landlords can create a property listing as a draft before publishing. This allows them to refine details over time without committing the listing publicly. They can review and finalize information, ensuring accuracy before making it visible to students.

13. **Student Reviews for Landlords**

After renting a property, students can leave feedback and rate their experience with the landlord. This information is visible to other students, promoting transparency and helping future renters make informed choices based on past tenant experiences.

14. **Saved Searches and Alerts**

Students can save specific search criteria and save it, improving their chances of finding a suitable rental without repeated manual searches.

15. **Bulk Image Upload for Listings**

Landlords can upload multiple images at once, allowing them to showcase different aspects of the property quickly and effectively. This feature ensures that listings are visually rich, providing students with a clearer view of the property.

16. **Property Listing Expiry Dates**

Listings have an automatic expiration period to ensure that only current rentals are displayed. Landlords receive notifications before expiry and can renew listings if the property is still available. This keeps the listing pool fresh and reduces outdated or inactive properties.

17. **Privacy Settings for User Profiles**

Users can control which profile details are publicly visible, such as hiding contact information from unverified users. This feature gives users control over their privacy, improving security and ensuring a safe environment for both students and landlords.

18. **In-Platform Document Exchange**

Students and landlords can securely upload and exchange necessary documents (e.g., rental agreements) directly on the platform. This keeps document handling organized and confidential, enhancing the rental process's security and convenience.

19. **Support for Multiple Property Listings per Landlord**

Landlords with more than one property can manage multiple listings under a single account, making it easy to track inquiries, update information, and maintain consistency across listings.

20. **Badges for Verified Landlords**

Verified landlords receive a badge on their profile, indicating they have passed a verification process. This badge helps students identify trusted landlords, promoting a safer and more reliable rental marketplace.

21. **Share Listings Externally**

Users can share listings via external channels, such as social media, email, or messaging apps, to increase visibility.

22. **Tenant Score Calculation Based on Questionnaire**

The system shall calculate a "Tenant Score" for each student based on responses to a landlord-provided questionnaire. Each question has a different assigned weight,

and the total score is determined by summing the scores from all responses. This Tenant Score provides landlords with an overall assessment to help them identify students best suited for their property.

23. **Display Nearby Amenities**

The system shall display nearby amenities such as cafes, transport hubs, grocery stores, and other points of interest when a student views a property's "Nearby Amenities" feature.

24. **Filter Amenities by Type**

The system shall allow students to filter the displayed amenities by type (e.g., restaurants, transportation, entertainment) to focus on specific facilities of interest around the property.

25. **User Account Creation (Admin, Moderator, Landlord, Student)**

The platform allows 4 user roles: Admin, Moderator Landlord, and Student. Admins have access to advanced platform management functions, including overseeing user roles, configuring platform settings, and handling security monitoring. Moderators will manage platform content and moderate listings; Landlords can list and manage properties; Students can browse, save, and rent properties. Each role has tailored permissions and access to specific features, ensuring a secure and organized experience.

26. **User Login and Logout**

Users can securely log in to access the platform’s features based on their role. Logging out protects account information, especially for students and landlords who may access sensitive messaging and profile data.

27. **Virtual Property Tours**

Landlords can include 360-degree virtual tours, allowing students to view properties remotely. This feature is especially useful for international students or those who cannot easily visit in person, giving them a realistic sense of the space.

28. **Video Call Request Between Student and Landlord**

The platform enables students and landlords to request and schedule video calls within the messaging system, allowing them to discuss rental details directly. Both parties can suggest or confirm time slots

5. List of Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions must render well on mobile devices
4. Data shall be stored in the database on the team's deployment cloud server.
5. Full resolution free media shall be downloadable directly, and full resolution media for selling shall be obtained after contacting the seller/owner
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users.
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application should follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used (optional for Fulda teams)
13. No email clients shall be allowed.
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today
17. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development
18. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0
19. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Fall 2024 For Demonstration Only" at the top of the WWW page. (Important to not confuse this with a real application).

6. Competitive Analysis

Features	WG	kleinanzeigen	Immo Scout24	Way2Home
Guest User	Yes	Yes	Yes	Yes
Browse Listings	Yes	Yes	Yes	Yes
Basic Search	Yes	Yes	Yes	Yes
Advanced Search Filters	Yes	Yes	Yes	Yes
Verified Listings	No	No	Yes	Yes
Favoriting Listing	Yes	Yes	Yes	Yes
In-App Messaging	Yes	Yes	Yes	Yes
Personalized Alerts	No	No	Yes	Yes
Tenant Score	No	No	No	Yes
Video Chat Communicatio n	No	No	No	Yes
Reviews & Ratings	No	No	Yes	Yes

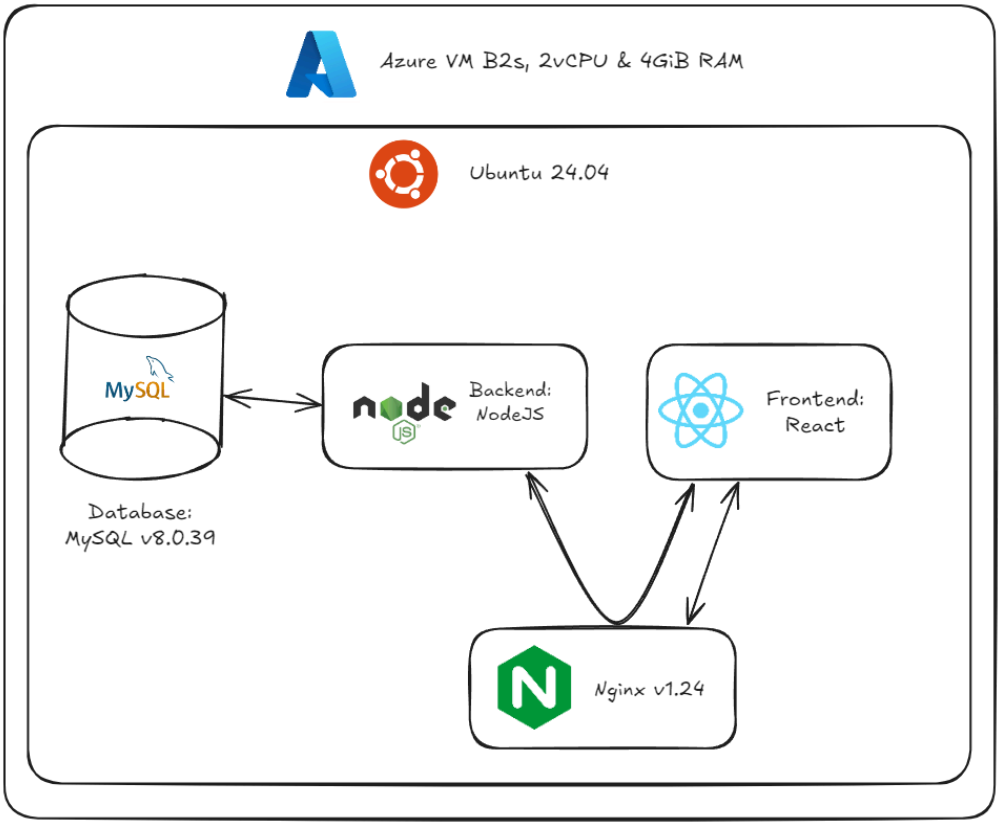
Way2Home provides all essential features expected in rental platforms, including guest browsing, listing favorites, in-app messaging, and both basic and advanced search filters, ensuring a smooth user search experience. **Way2Home** distinguishes itself from competitors with key student-centric features not offered by WG-Gesucht, eBay Kleinanzeigen, or ImmobilienScout24. While ImmobilienScout24 includes verified listings, Way2Home enhances this security with both verified listings and **tenant scoring** to give landlords an idea of tenant reliability. Unlike other platforms, Way2Home integrates video chat communication, making it easier for international

and remote students to interact directly with landlords, adding convenience and security. Furthermore, the reviews and ratings feature helps build transparency and trust in the rental process. With these unique features, Way2Home creates a tailored, reliable, and student-focused experience, making it a strong choice for student accommodation in Germany.

7. High-Level System Architecture and Technologies Used

Technologies

- Server host: Azure VM B2s, 2vCPUs 4 GiB RAM
- OS: Ubuntu 24.04
- Database: MySQL v8.0.39
- Web server: Nginx v1.24.0
- Server side language: Node.js
- Web framework: React
- IDE: Visual Studio Code
- SSL Cert: Let's Encrypt (certbot v2.9.0)



Deployment Information

Supported browsers are Chrome 130 & Firefox 132.

Deployment is done via GitHub actions. The action builds the software, moves it to the server via SCP, and then restarts the service to serve the new version.

8. Team and Roles

Name	**Role**	**Responsibilities**
Romina Nikasa	Team Lead	Keep the team focussed on the goals, communication with Professor
Rashid Hussain Mohammed	GitHub Master	Organizes Git and makes sure commit policies are followed
Tim Schröder	Backend Lead	Review backend PRs
Anmoldeep Angadi	Frontend Lead	Review frontend PRs
Samarth Mahajan	Frontend Developer	Frontend side Development
Suhaila Kondappilly Aliyar	Backend Developer	Backend side Development
Maira Shafi	Documentation Master	Maintain Documentation, and overall Accuracy and Relevance

9. Checklist

Review Tasks: Answer the following items with either 'DONE,' 'ON TRACK,' or 'ISSUE':

Meeting schedule established	DONE
GitHub master chosen	DONE
Agreement on software tools	DONE
Readiness to use front-end/back-end frameworks	ON TRACK
Team understanding of the final M1 document	DONE
Organization of the GitHub repository	ON TRACK